Name _____ Period _____ Date _____

7-3 Lesson Quiz: Graphing Sine and Cosine Functions

short form

Question numbers match the Savvas Lesson Quiz; questions 3 and 5 are not on this form.

1. Select **all** the correct statements.

☐ **A.** The amplitude of $y = 2\cos\left(\frac{1}{2}x\right)$ is $\frac{1}{2}$.

☐ **B.** The period of $y = 2\cos\left(\frac{1}{2}x\right)$ is π.

☐ **C.** The amplitude of $y = \frac{\pi}{2}\sin\left(\frac{\pi}{2}x\right)$ is $\frac{\pi}{2}$.

☐ **D.** The period of $y = 4\pi\sin\left(\frac{1}{2}x\right)$ is 4π.

☐ **E.** The period of $y = \cos(2\pi x)$ is 1.

☐ **F.** The amplitude of $y = 2\sin\left(\frac{1}{2}x\right)$ is 2.

Show one period calculation: _____

2. What is the average rate of change of the function $y = 3\sin(2x)$ on the interval $\left[0, \frac{\pi}{4}\right]$?

Ⓐ $-\frac{12}{\pi}$ Ⓑ $-\frac{6}{\pi}$ Ⓒ $\frac{6}{\pi}$ Ⓓ $\frac{12}{\pi}$

Show your work: the values of y at $x = 0$ and at $x = \frac{\pi}{4}$, and the rate of change.

4. The graph of the function $f(x) = 5\cos(2x)$ is a reflection over the x-axis of the graph of

$$g(x) = _____\ _____(_____).$$

Write one tile in each blank. | -5 | | cos | | $(2x)$ | | sin | | $(-2x)$ | | 5 |

Scoring: Question 1 and Question 2 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20.

7-3 Lesson Quiz, short form: answer key

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1. **C, D, E, F.** Amplitude $= |a|$, period $= \frac{2\pi}{|b|}$. A is false: the amplitude of $2\cos\left(\frac{1}{2}x\right)$ is 2 ($\frac{1}{2}$ is b, not a). B is false: its period is $\frac{2\pi}{1/2} = 4\pi$ (B multiplies 2π by $\frac{1}{2}$, the Try It 3 common error). C: $\left|\frac{\pi}{2}\right| = \frac{\pi}{2}$. D: $\frac{2\pi}{1/2} = 4\pi$ (the 4π in front is the amplitude and does not change the period). E: $\frac{2\pi}{2\pi} = 1$. F: $|2| = 2$. Savvas quiz item 1, DOK 1, Skills Review and Practice F90. Taught by Example 3 (Try It 3 is the same two formulas). **10 points:** 2.5 for each of C, D, E, F checked; subtract 2.5 for each of A, B checked; minimum 0.

2. **D, $\frac{12}{\pi}$.** The period of $3\sin(2x)$ is $\frac{2\pi}{2} = \pi$, so the key points are $\frac{\pi}{4}$ apart: $(0, 0)$ and the maximum $\left(\frac{\pi}{4}, 3\right)$. Average rate of change $= \dfrac{3\sin\left(\frac{\pi}{2}\right) - 3\sin 0}{\frac{\pi}{4} - 0} = \dfrac{3}{\pi/4} = \dfrac{12}{\pi}$. A has the wrong sign (the graph rises from 0 to the maximum); C ($\frac{6}{\pi}$, that is $\frac{3}{\pi/2}$) divides by the half period $\frac{\pi}{2}$ instead of $\frac{\pi}{4}$, and B has that error with the wrong sign. Savvas quiz item 2, DOK 1, F31. Taught by Example 4 (the five key points, a zero then a maximum a quarter period later) and the packet's Practice (the same rate of change on Example 4's graphs); the rate formula is the Topic 1 skill. **10 points:** 3 for $y(0) = 0$ and $y\left(\frac{\pi}{4}\right) = 3$, 3 for the quotient set up over $\frac{\pi}{4} - 0$, 4 for choice D.

4. **-5; cos; $(2x)$, so $g(x) = -5\cos(2x)$.** Reflecting over the x-axis replaces y with $-y$: $-\big(5\cos(2x)\big) = -5\cos(2x)$, and reflecting g back gives f. Also correct: -5; cos; $(-2x)$, because cosine is even, so $-5\cos(-2x) = -5\cos(2x)$; give it full credit. The tiles 5, cos, $(-2x)$ are a trap: $5\cos(-2x) = 5\cos(2x)$ is f itself, not its reflection. Savvas quiz item 4, DOK 1, F90. Taught by Example 3B ($y = -f(x)$ is a reflection over the x-axis), rehearsed by Try It 4b ($h(x) = -\sin 2\pi x$) and the packet's extension (Practice 23, Part E). **Bonus 5:** 3 for -5, 1 for cos, 1 for $(2x)$ or $(-2x)$.

Items left off this form (available for a retake or Form B): Savvas item 3, the equation $f(x) = 2\cos\left(\frac{\pi}{2}x\right)$ against the graph of g (a sine curve through the origin with maxima at $\left(\frac{1}{2}, 2\right)$ and $\left(\frac{5}{2}, 2\right)$): how do the period and the amplitude compare? (answer B: the amplitudes are both 2, and the period of f, $\frac{2\pi}{\pi/2} = 4$, is twice the period of g, $\frac{5}{2} - \frac{1}{2} = 2$; DOK 2, F90; the cut Example 6); Savvas item 5, the amplitude of the tide model with high tide 5 ft and low tide 1 ft (answer 2 ft, $\frac{5-1}{2}$; DOK 2, F90; the cut Example 5, and Practice 20 is its DOK 3 cousin). Both are left off because no kept Example teaches them. Item 5 is a one-round whiteboard bridge if the teacher puts it back: amplitude $= \frac{\max - \min}{2}$, midline $= \frac{\max + \min}{2}$.

IXL replacement: SmartScore 70 or above on 9S8 and A7V replaces this quiz with 20/20.